Hourly Temperature Prediction Using NOAA Data

A standalone comparison of Multiple Linear Regression, Gradient Boosting, and an LSTM recurrent neural network for 15-day hourly temperature prediction using common NOAA variables.

Dataset	136,505 hourly timestamps
Period	1 January 2011 to 28 July 2026
Scope	Temperature prediction using common NOAA weather variables
Final test	360 consecutive hours; 359 observed targets
Forecast	28 July 2026 17:00 to 12 August 2026 16:00 UTC

Best final-test RMSE: LSTM Recurrent Neural Network (2.755 C). Best seasonal-validation RMSE: Histogram Gradient Boosting (4.492 C).

The evaluation is deliberately recursive: once each 15-day prediction begins, none of the actual future temperature values are supplied to the model. This makes the reported errors representative of the requested forecasting task.

1. Dataset and preparation

The source is NOAA's Global Historical Climatology Network hourly (GHCNh), station GRMU0016622, THESSALONIKI GREECE (MIKRA), at 40.52 N, 22.97 E and 4 m elevation. Sixteen station-year files covering 2011-2026 contained 314,025 raw reports. Multiple observations within an hour were consolidated into one UTC hourly record.

Cleaning rules

Step	Treatment
Quality control	Applied NOAA source-specific quality flags and physical-range checks.
Hourly aggregation	Numeric weather observations were averaged; wind direction used a circular mean.
Precipitation	The final report in each hour was retained, following NOAA guidance.
Missing timestamps	A complete hourly UTC index was retained; missing measurements remain explicitly blank.
Temperature coverage	99.64% of the 136,505 hours.

Modelling variables

Temperature is the prediction target. Predictor histories include temperature, dew point, relative humidity, wind speed and direction, visibility, and altimeter pressure. Calendar cycles, lags from 1 to 168 hours, and rolling summaries are produced by the code. Sparse wind-gust, precipitation, ceiling, and station-pressure fields remain in the source CSV for completeness but are excluded from model fitting.

The modelling scope uses common, consistently available NOAA weather variables. PM10 and pyranometer measurements are not supplied by GHCNh and are therefore excluded. Temperature is the sole prediction target.

Source

NOAA/NCEI GHCNh product: https://www.ncei.noaa.gov/products/global-historical-climatology-network-hourly

2. Evaluation design

A random 70/30 split was not used because it would mix past and future observations. Instead, all divisions are chronological and every model receives the same prediction origins.

Stage	Dates and purpose
Development training	Data before 1 January 2024; used to fit the validation model.
Seasonal validation	Eight separate 15-day windows across January, April, July and October in 2024-2025.
Final test training	All available observations before 13 July 2026 17:00 UTC.
Final test	13 July 2026 17:00 to 28 July 2026 16:00 UTC; 360 hourly predictions.
Future forecast	Models retrained through the latest observation, then forecast 360 hours ahead.

Recursive forecasting

Predicted temperature is fed back into subsequent lag and rolling features. Future non-target weather observations are not used. Their short-term paths follow a transparent weekly-persistence assumption based on the same hour seven days earlier. This represents one uninterrupted 15-day forecast from a single starting point.

Models

Model	Implementation
Multiple Linear Regression	Ordinary least-squares regression over 47 calendar, lag and rolling features.
Gradient Boosting	Histogram gradient-boosted regression trees with early stopping and regularisation.
LSTM/RNN	A recurrent network with LSTM gates, a 24-hour sequence, 16 hidden units and 12 epochs.

Metrics

MAE is the average absolute error; lower is better. MSE averages squared errors and penalises large misses; lower is better. RMSE is the square root of MSE in degrees Celsius; lower is better. R2 measures explained variation; 1 is ideal, 0 matches a mean-value baseline, and a negative value is worse than that baseline. Mean relative error expresses absolute error as a percentage of the actual temperature magnitude; lower is better.

Relative error is interpreted cautiously because temperatures close to 0 C create very large percentages from modest absolute errors. MAE, RMSE and R2 are therefore the primary comparison metrics.

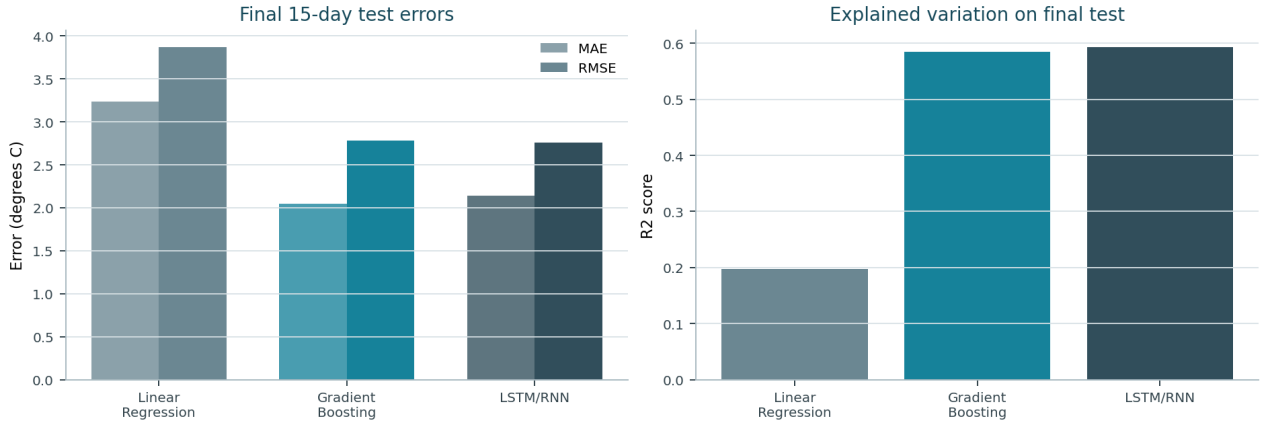
3. Results

Seasonal validation

Model	MAE (C)	MSE	RMSE (C)	R2	Mean relative error
Multiple Linear Regression	4.151	26.277	5.126	0.675	37.72%
Histogram Gradient Boosting	3.628	20.174	4.492	0.750	46.93%
LSTM Recurrent Neural Network	3.622	20.420	4.519	0.747	48.39%

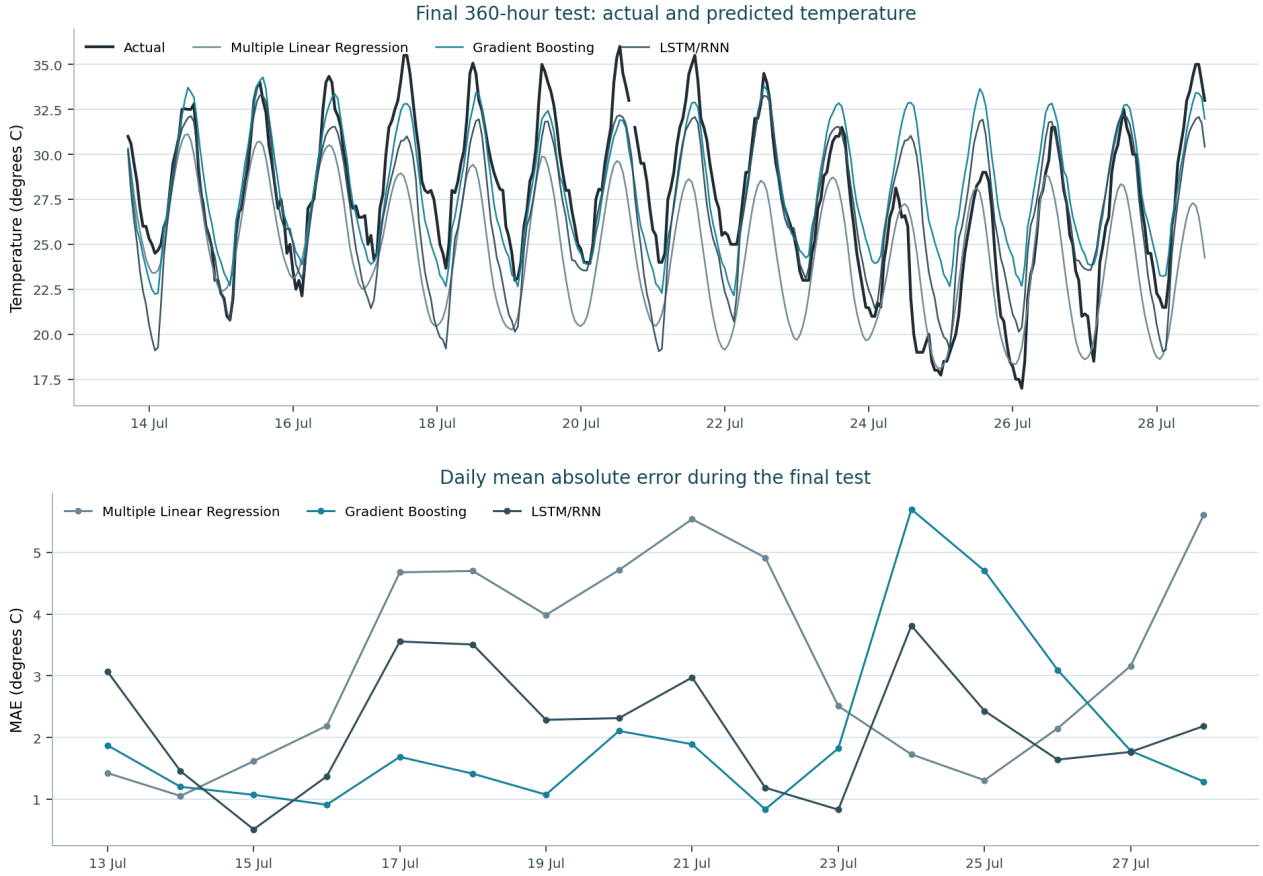
Final 15-day test

Model	MAE (C)	MSE	RMSE (C)	R2	Mean relative error
Multiple Linear Regression	3.238	14.995	3.872	0.197	11.47%
Histogram Gradient Boosting	2.049	7.753	2.784	0.585	8.37%
LSTM Recurrent Neural Network	2.138	7.592	2.755	0.594	8.25%



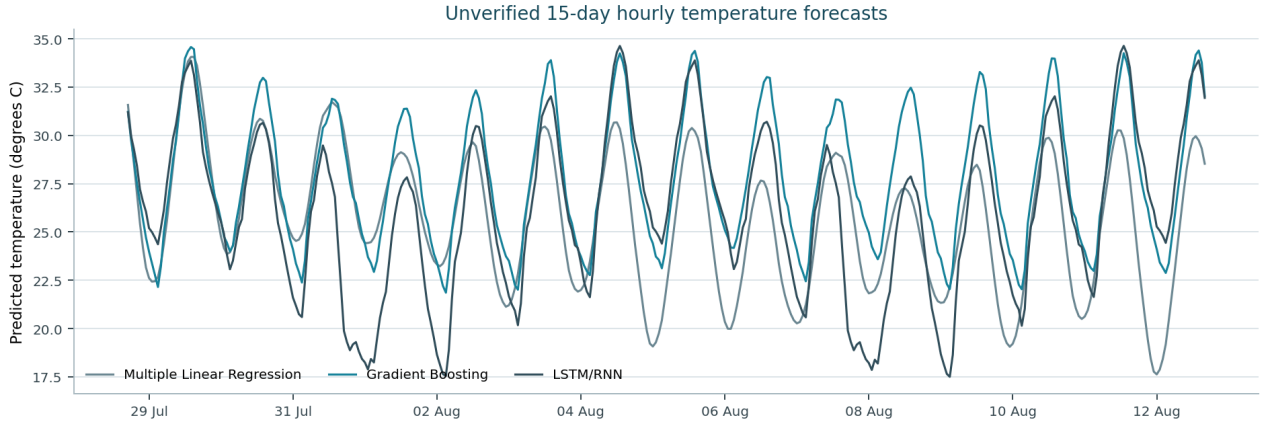
LSTM Recurrent Neural Network produced the lowest final-test RMSE. Multiple Linear Regression achieved a positive R2, but its higher MAE and RMSE show that it remained less accurate than the two nonlinear models over this 15-day period.

4. Prediction behaviour



The plots show how errors accumulate across the continuous forecast rather than presenting disconnected one-hour predictions. A model can follow the daily cycle while still missing the level or amplitude of a warm or cool period, which increases multi-day error.

5. Fifteen-day forecast and conclusion



These values are model forecasts beyond the last NOAA observation and are not verified against actual future measurements. The historical NOAA series and final-test predictions remain in task9_noaa_data_and_predictions.csv. The separate task9_15_day_forecast.csv contains the 360 future hourly forecasts from 28 July 2026 17:00 through 12 August 2026 16:00.

Conclusion

The NOAA dataset provides a long, consistent hourly record for comparing all three methods under the same conditions. LSTM Recurrent Neural Network was the strongest model on the final 15-day test. The final selection should also consider the eight-window seasonal validation because one July test period cannot represent all annual weather conditions.

Limitations

The data come from one station. Missing weather predictors are filled using training-period medians after short forward fills. Exogenous weather conditions during recursive prediction follow weekly persistence rather than an independent numerical weather forecast. PM10 is outside the scope because it is absent from NOAA GHCNh.